5. Atomic Physics

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CONTENTS

5.1 THE NUCLEAR ATOM

5.1.1 ATOMIC MODEL

The Structure of the Atom

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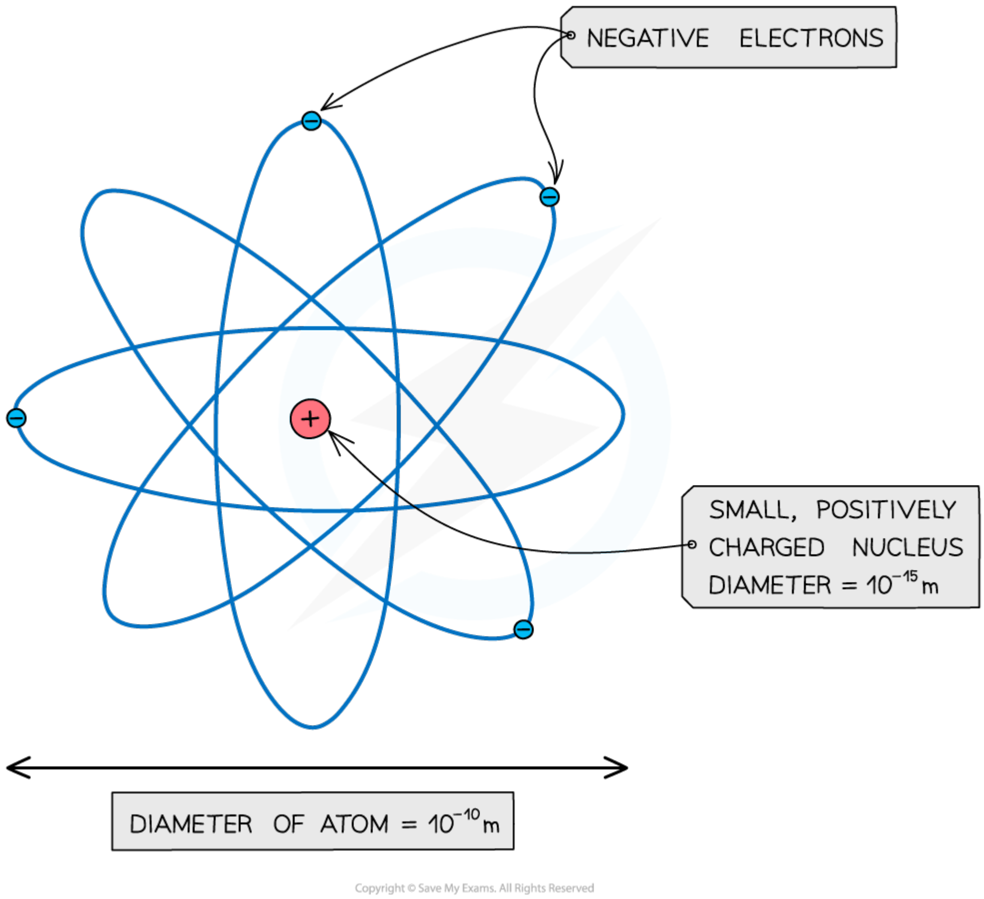
- Atoms consist of small dense positively charged nuclei, surrounded by negatively charged electrons



An atom: a small positive nucleus, surrounded by negative electrons

(Note: the atom is around 100,000 times larger than the nucleus!)

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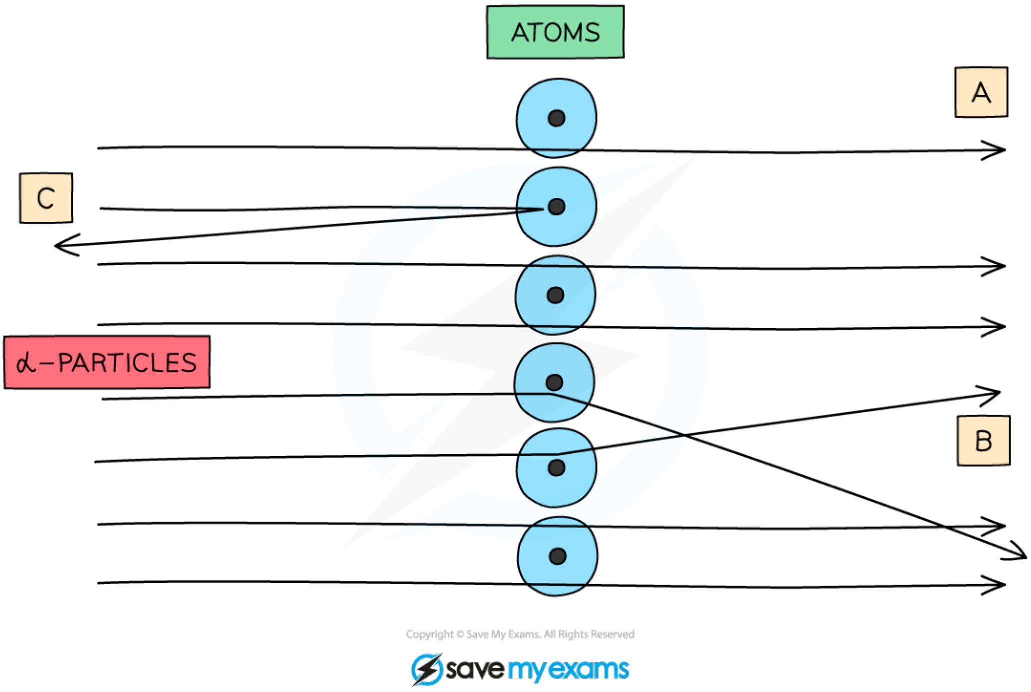
Alpha Particle Scattering

- Evidence for the structure of the atom comes from the study of α-particle scattering



When α-particles are fired at thin gold foil, most of them go straight through but a very small number bounce straight back

- When α-particles are fired at thin pieces of gold foil:
  - **The majority of them go straight through (A)**
    This happens because the atom is mainly empty space
  - **Some are deflected through small angles (B)**
    This happens because the positive α-particles are repelled by the positive nucleus
  - **A very small number are deflected straight back (C)**
    This is because the nucleus is extremely small

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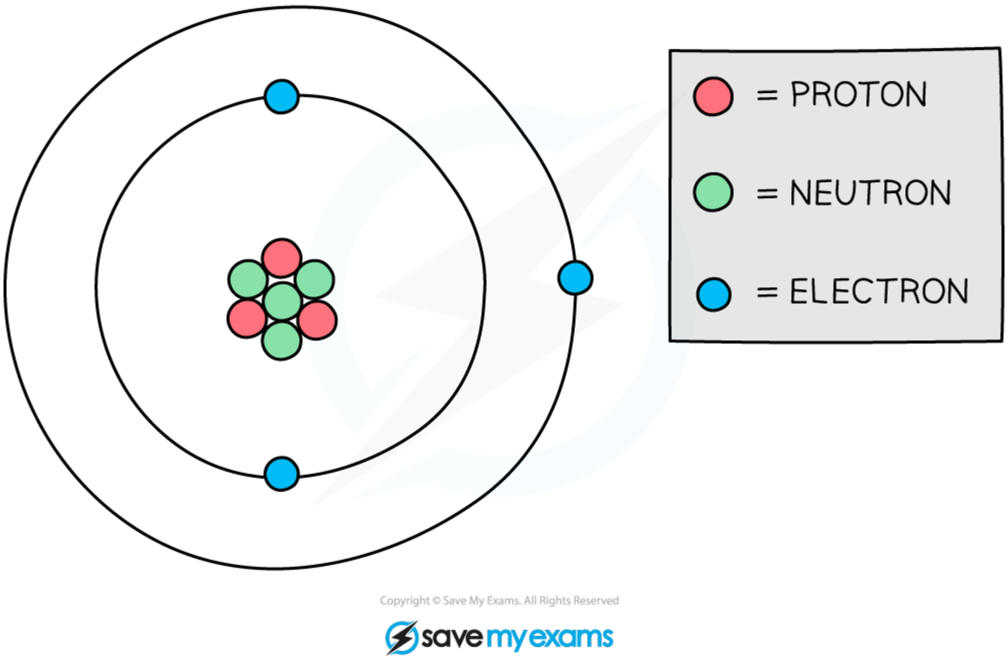
5.1.2 NUCLEUS

Protons & Neutrons

- Atoms are made up of three different particles:



Protons and neutrons are found in the nucleus of an atom

The properties of each of these particles is shown in the table below:

PARTICLE	RELATIVE CHARGE	RELATIVE MASS
PROTON	+1	1
NEUTRON	0	1
ELECTRON	−1	1/2000 (NEGLIGIBLE)

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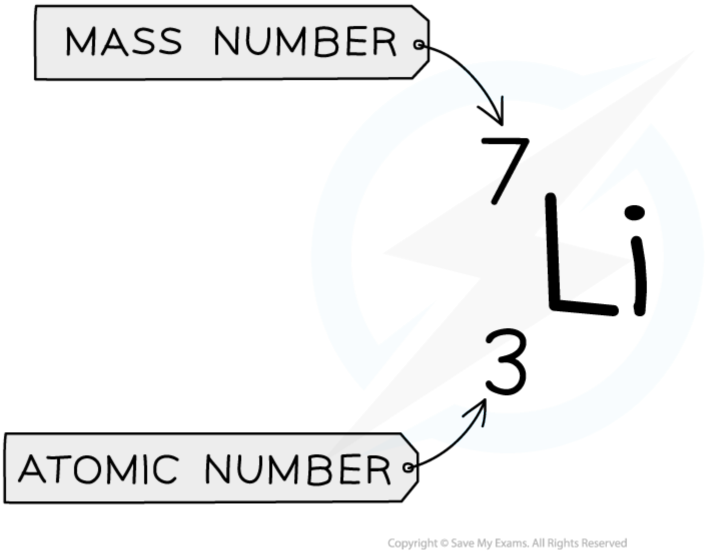
- The atom shown in the above diagram can also be represented using an atomic symbol:



Atomic symbols, like the one above, describe the constituents of nuclei

- The top number is called the **nucleon number, A**, and is equal to the total number of particles (protons and neutrons) in the nucleus
- The lower number is called the **proton number, Z,** and is equal to the total number of protons in the nucleus

  (Note: Chemists refer to the nucleon number as the **mass number**, and the proton number as the **atomic number**)

- When given an atomic symbol, you can figure out the total number of protons, neutrons and electrons in the atom:
  - **Protons:** The number of protons is equal to the proton number
  - **Electrons**: Atoms are neutral, and so in a neutral atom the number of negative electrons must be equal to the number of positive protons
  - **Neutrons**: The number of neutrons can be found by subtracting the proton number from the nucleon number
- The term **nucleon** is used to mean a particle in the nucleus – ie. either a proton or a neutron
- The term **nuclide** is used to refer to a nucleus with a specific combination of protons and neutrons

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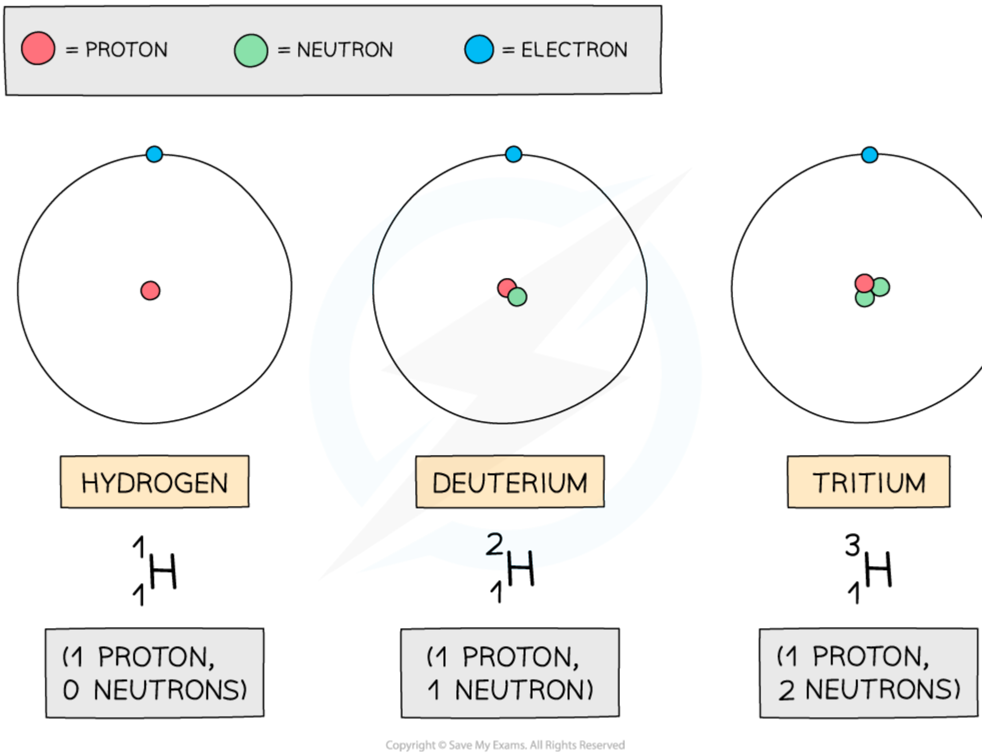
Isotopes

- Although all atoms of the same element always have the same number of protons (and hence electrons), the number of neutrons can vary:



The three atoms shown above are all forms of hydrogen, but they each have different numbers of neutrons

- The number of neutrons in an atom does not affect the chemical properties of an atom, only its mass. Such atoms are called isotopes:
  Isotopes are atoms (of the same element) that have equal numbers of protons but different numbers of neutrons

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5.1.3 NUCLEAR REACTIONS

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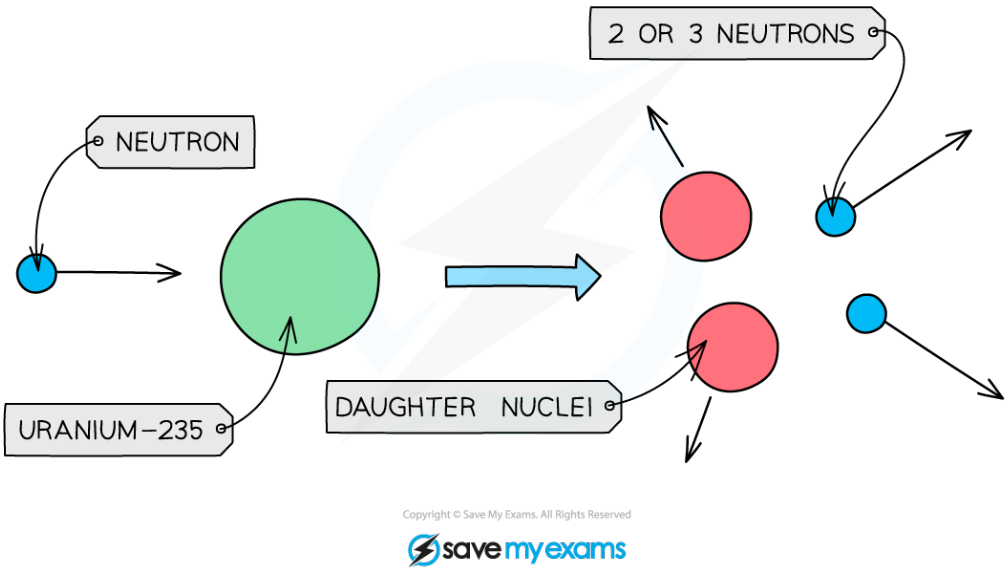
Nuclear Fission

- Usually large unstable nuclei break up gradually by the process of radioactive decay, but a small number (including Uranium-235, a naturally occurring isotope of Uranium) can break up in one big go – a process known as **nuclear fission**
- In order to undergo nuclear fission, a nucleus usually requires some energy which can be given by hitting the nucleus with a neutron
  (Neutrons are used because they are chargeless and so are not repelled by the positive charge of the nucleus)



Nuclear fission: A Uranium-235 nucleus is struck by a neutron, breaking it into two smaller daughter nuclei and 2 or 3 neutrons

- When this happens, the original nucleus breaks apart into **two smaller 'daughter' nuclei, along with two or three neutrons**
- These **fission products** carry away the energy released in the form of kinetic energy

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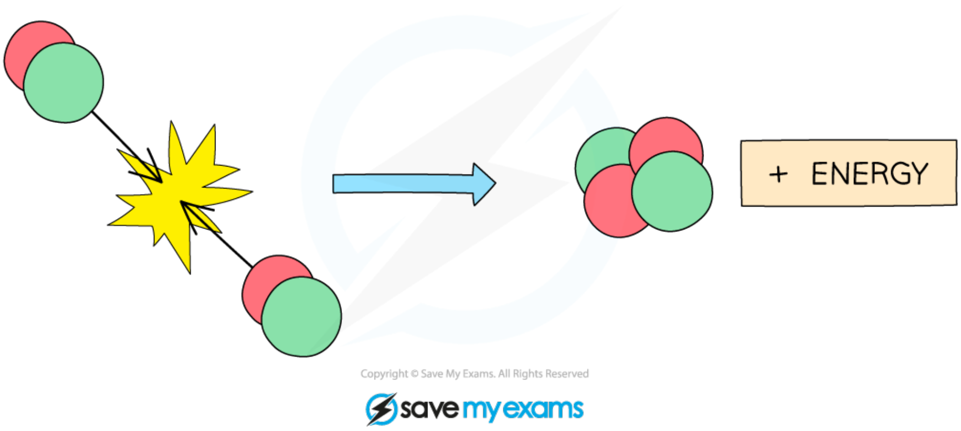
Nuclear Fusion

- Nuclear fusion involves taking small nuclei (such as hydrogen) and colliding them together at high speed to form larger nuclei



Fusion is the process in which small nuclei, such as hydrogen, are fused together to form larger nuclei

- This process also releases energy

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Nuclear Equations

- Nuclear reactions, such as fission and fusion, can be represented using nuclear equations (which are similar to chemical equations in Chemistry)
  For example:

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{90}_{38}Sr + ^{144}_{54}Xe + 2^{1}_{0}n$$

- The above equation represents a fission reaction in which a Uranium nucleus is hit with a neutron and splits into two smaller nuclei – a Strontium nucleus and a Xenon nucleus, releasing two neutrons in the process
- In the above reaction:
  The sum of top (nucleon) numbers on the left-hand side equals the sum of top number on the right-hand side:

$235 + 1 = 236 = 90 + 144 + 2 \times 1$

The same is true for the lower (proton) numbers:

$92 + 0 = 92 = 38 + 54 + 2 \times 0$

- By balancing equations in this way, you can determine, for example, the number of neutrons emitted by a process like this

  Example:

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{96}_{37}Rb + ^{137}_{55}Cs + N^{1}_{0}n$$

- In the above example, balancing the numbers on the top shows that 3 neutrons must be released in the reaction (i.e. *N = 3*)

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Exam Question: Easy

Hydrogen is an element with one proton. It has three different isotopes.

Which statement about the three Hydrogen isotopes is correct?

A They must have different numbers of electrons orbiting their nucleus.

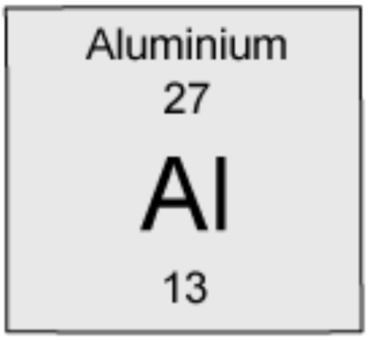
B They must all have one proton in their nuclei.

C They must all have one neutron in their nuclei.

D They must all have the same number of nucleons in their nucleus.

Exam Question: Hard

How many neutrons are in a nucleus of Aluminium? The chemical symbol is given below:



A 13

B 27

C 14

D 40

> CHECK YOUR ANSWERS AT SAVEMYEXAMS.CO.UK

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5.2 RADIOACTIVITY

5.2.1 DETECTION OF RADIOACTIVITY

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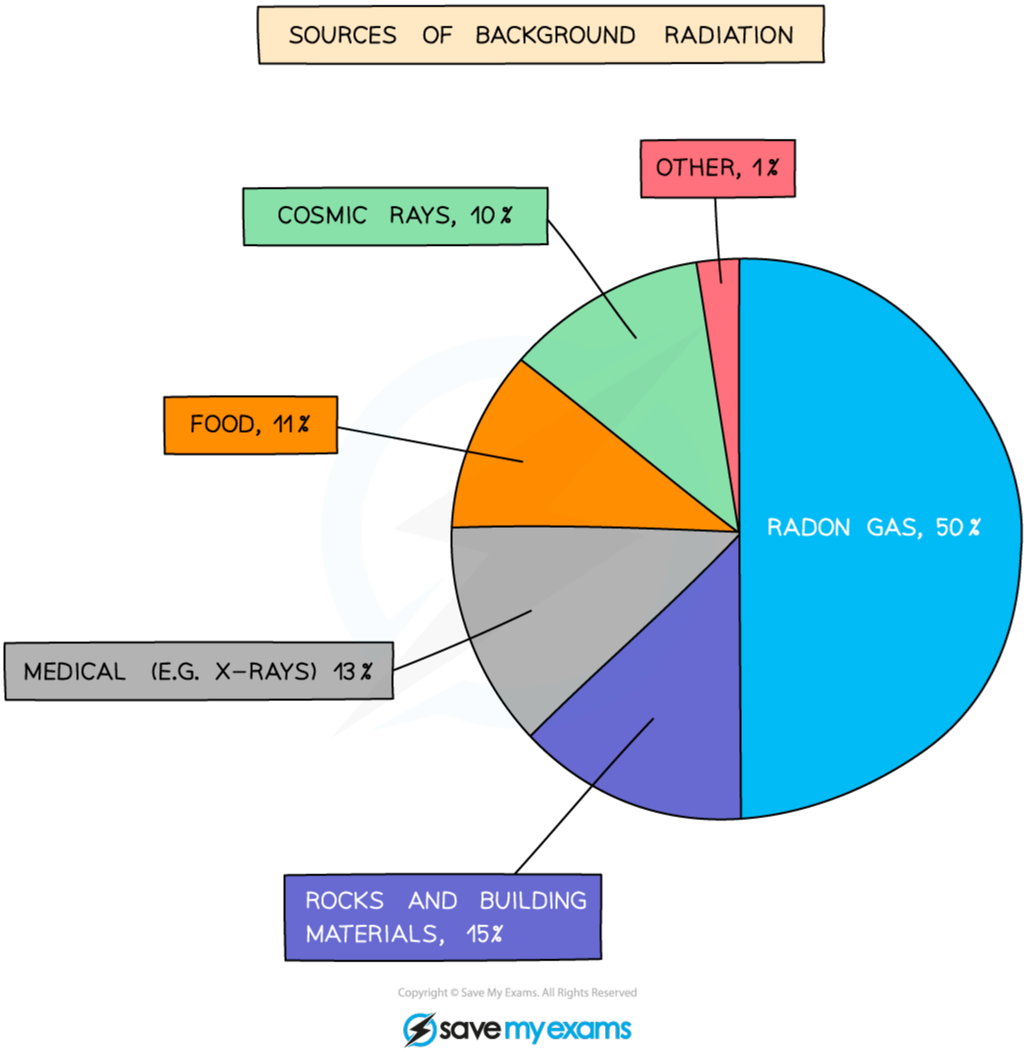
Background Radiation: Basics

- Background radiation is the radiation that is always present around us in the environment



Background radiation is the radiation that is present all around in the environment

- Although most background radiation is natural, a small amount of it comes from artificial sources, such as **medical procedures** (including X-rays)
  (Radiation from Nuclear Power come to less than 0.1% of the total)
- Levels of background radiation can vary significantly from place to place

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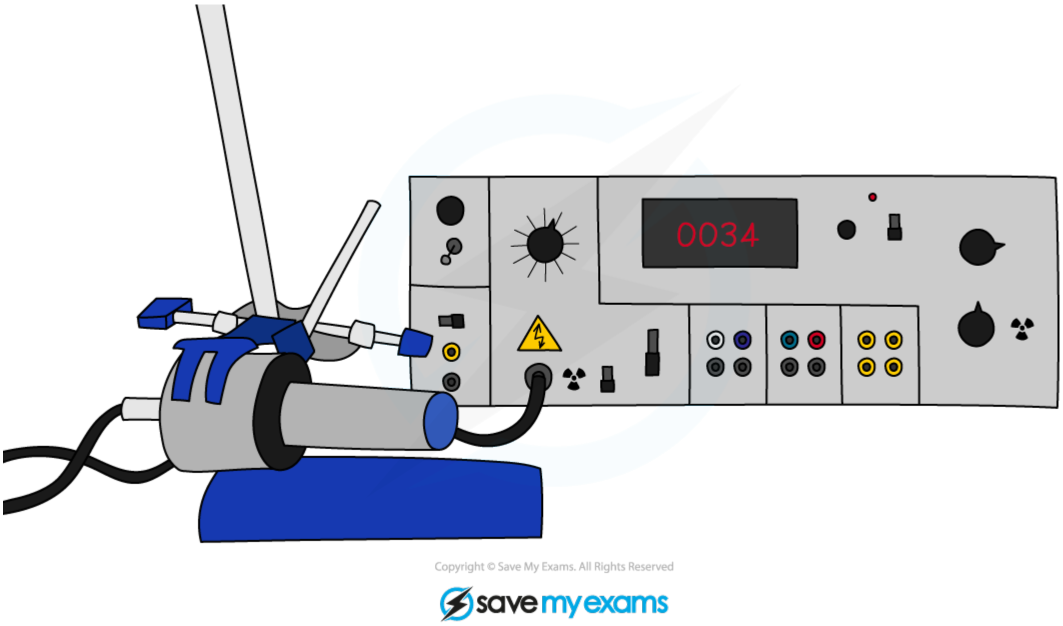
Detecting Radiation

- When radiation passes close to an atom, it knocks out electrons, ionising the atom
- Radiation detectors work by detecting the presence of these ions or the chemical changes that they produce
- Examples of radiation detectors include:
  - **Photographic film** (often used in badges)
  - **Geiger-Muller** (GM) tubes
  - **Ionisation chambers**
  - **Scintillation counters**
  - **Spark counters**



A Geiger-Muller tube (or Geiger counter) is a common type of radiation detector

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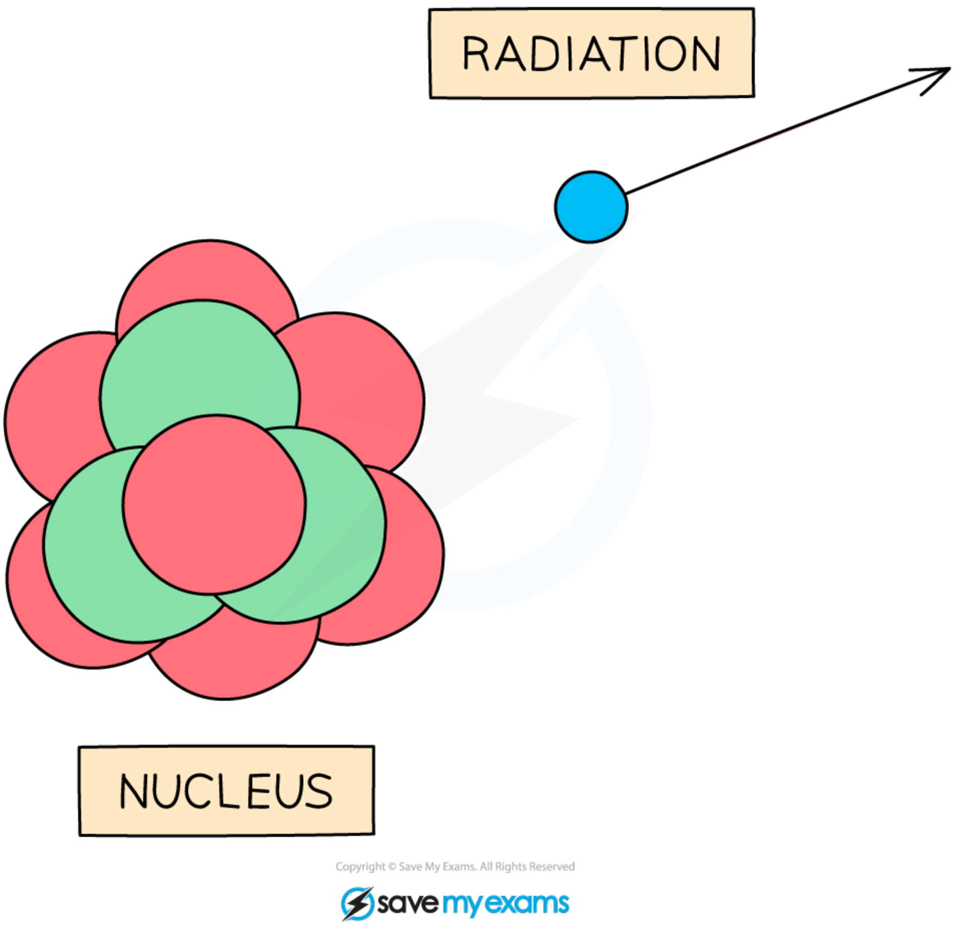
5.2.2 CHARACTERISTICS OF RADIATION

The Nature of Decay

- Radiation consists of high energy particles (or waves) emitted from the nucleus of an unstable atom



Unstable nuclei decay by emitting high energy particles – radiation

- There are three (main) types of radiation: **alpha (α) particles, beta (β^-) particles, or gamma (γ) rays**
- **Radiation is emitted randomly**
  This means that, although we understand *why* some nuclei emit radiation, it is **impossible to predict exactly when** a nucleus will emit radiation

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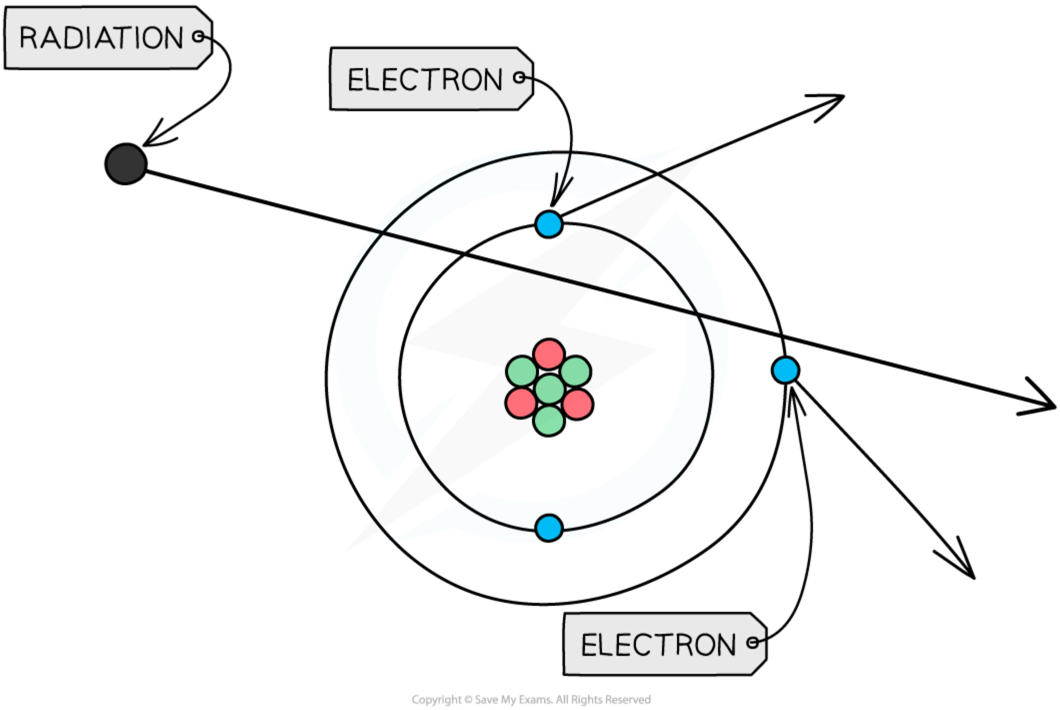
The Properties of Radiation

- **Alpha (α) particles** are high energy particles made up of **2 protons and 2 neutrons** (the same as a helium nucleus).
  They are usually emitted from nuclei that are too large
- **Beta (β^-) particles are high energy electrons** emitted from the nucleus (even though the nucleus does not normally contain any electrons)
  They are usually emitted by nuclei that have too many neutrons
- **Gamma (γ) rays are high energy electromagnetic waves**
  They are emitted by nuclei that need to lose some energy
- If these particles hit other atoms, they can knock out electrons, **ionising the atom**



When radiation passes close to atoms, it can knock out electrons, ionising the atom

- Ionisation can cause chemical changes in materials, and can damage or kill living cells

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The nature and properties of the different types of radiation are summarised in the table below

PARTICLE	WHAT IS IT	CHARGE	RANGE IN AIR	PENETRATION	IONISATION
ALPHA (α)	2 PROTONS + 2 NEUTRONS	+2	FEW cm	STOPPED BY PAPER	HIGH
BETA (β^-)	ELECTRON	−1	FEW 10s OF cm	STOPPED BY FEW MM ALUMINIUM	MEDIUM
GAMMA (γ)	ELECTROMAGNETIC WAVE	0	INFINITE	REDUCED BY FEW MM LEAD	LOW

- Note that when you go down the table, **the range and penetration increase, but the ionisation decreases**

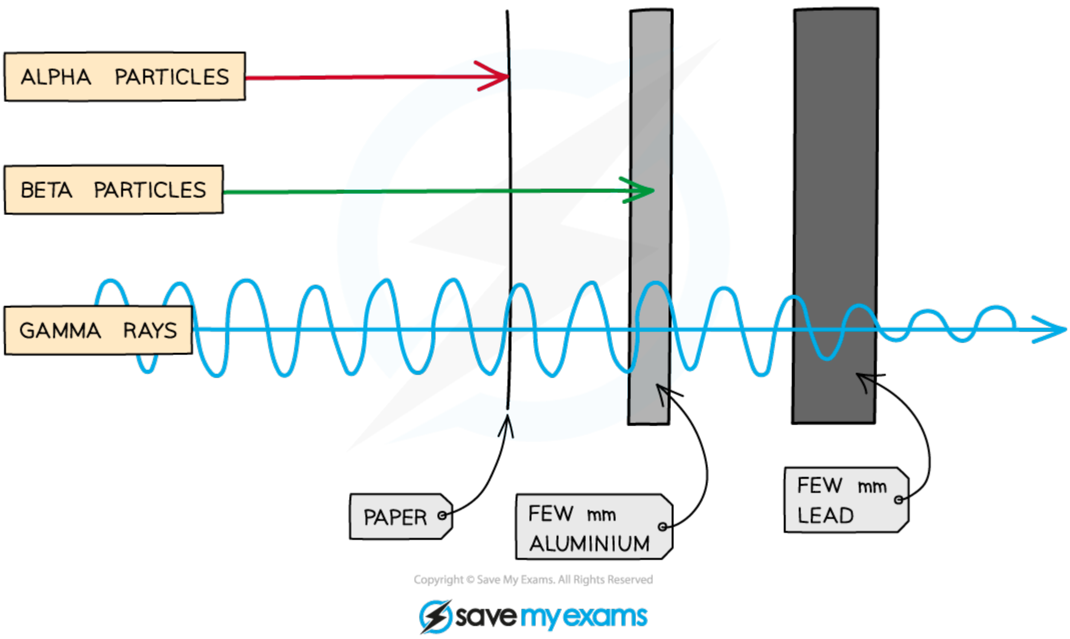

Diagram showing the penetrative abilities of alpha (α) particles, beta (β⁻) particles, and gamma (γ) rays. Note how some of the gamma rays are able to penetrate the lead

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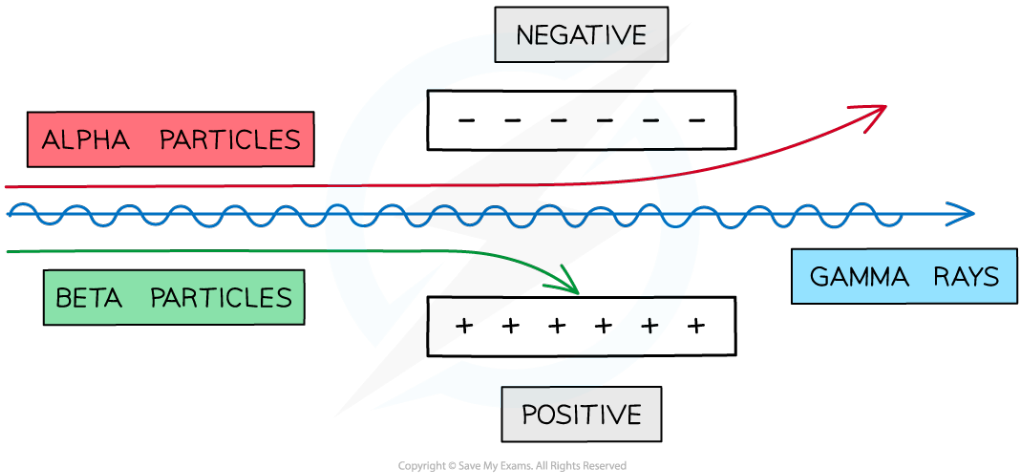
Deflection in Electric & Magnetic Fields

- Because of their charges, alpha and beta particles can be deflected by electric and magnetic fields



Alpha and Beta particles can be deflected by electric fields

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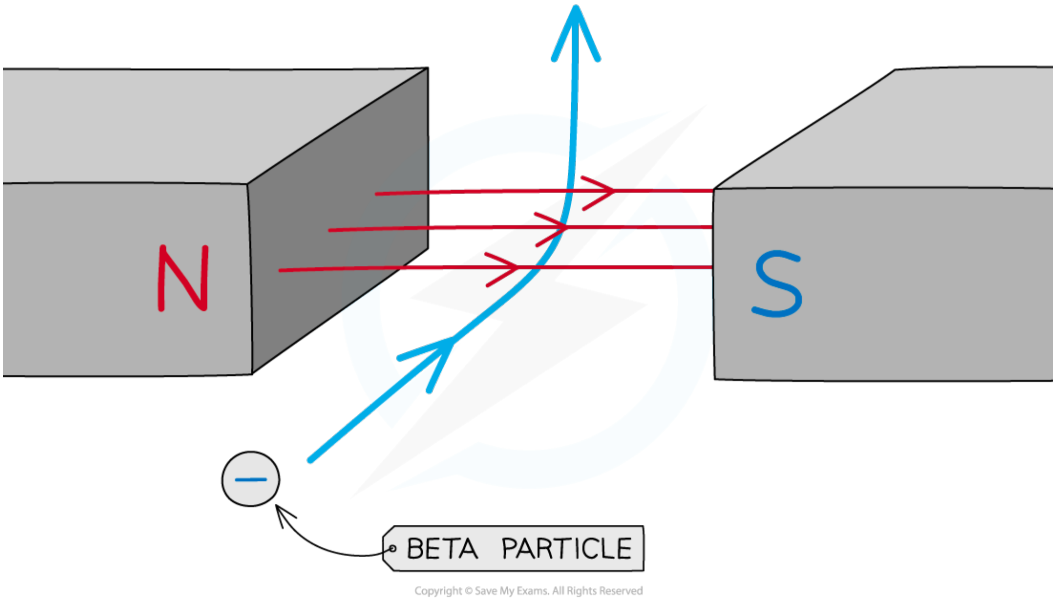
- Because they have opposite charges, alpha and beta particles are deflected in opposite directions
- Beta is deflected by more than alpha, because beta particles have a much smaller mass
- Gamma is not deflected because gamma rays have no charge



Alpha and Beta particles can also be deflected by magnetic fields

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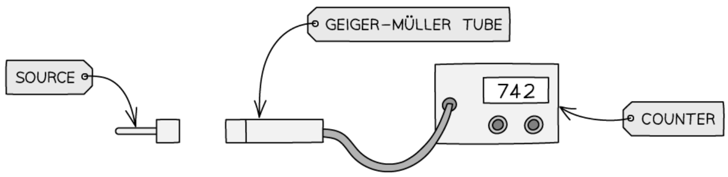
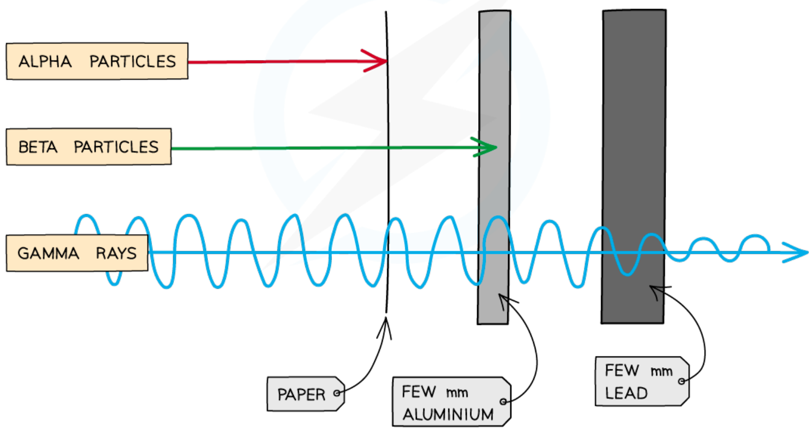
WORKED EXAMPLE: A RADIOACTIVE SOURCE IS PLACED IN FRONT OF A GM TUBE AND RECORDS 742 COUNTS PER MINUTE. WHICH TYPES OF RADIATION DOES THE SOURCE EMIT?



MATERIAL OF SHEET BETWEEN SOURCE AND TUBE	THIN CARD	ALUMINIUM FOIL	THICKED LEAD
AVERAGE COUNT RATE/COUNTS PER MINUTE	273	275	68



ALPHA PARTICLES ARE STOPPED BY PAPER. THE TABLE SHOWS THE COUNT REDUCES FROM 742 TO 273 SO THE SOURCE MUST EMIT ALPHA PARTICLES

BETA PARTICLES ARE STOPPED BY ALUMINIUM FOIL. THE TABLE SHOWS THE COUNT DOES NOT REDUCE SIGNIFICANTLY SO THE SOURCE DOES NOT EMIT BETA PARTICLES

GAMMA RAYS ARE STOPPED BY THICK LEAD. THE TABLE SHOWS 273 TO 68 SO THE SOURCE MUST EMIT GAMMA RAYS

Diagram showing an experiment to find the type of radiation being emitted by a source

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Ionisation

- **Alpha** is by far the most ionising form of radiation
    - Alpha particles leave a dense trail of ions behind them, affecting virtually every atom they meet
    - Because of this they quickly lose their energy and so have a short range
    - Their short range makes them relatively harmless if handled carefully, but they have the potential to be extremely dangerous if the alpha emitter enters the body
- **Beta** particles are moderately ionising
    - The particles create a less dense trail of ions than alpha, and consequently have a longer range
    - They tend to be more dangerous than alpha because they are able to travel further and penetrate the skin, and yet are still ionising enough to cause significant damage
- **Gamma** is the least ionising form of radiation (although it is still dangerous)
    - Because Gamma rays don't produce as many ions as alpha or beta, they are more penetrating and have a greater range
    - This can make them hazardous in large amounts

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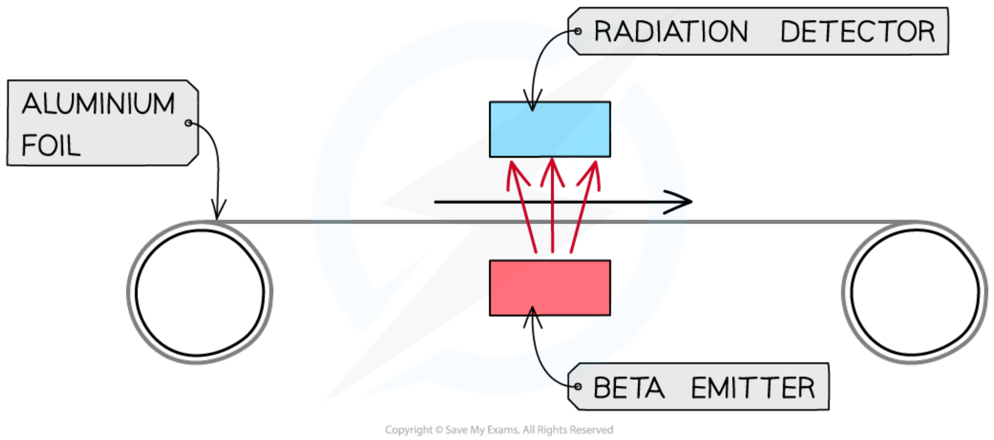
Applications of Radioactivity

- Radioactivity has a large number of uses in both medicine and industry, some of which are listed below

Measuring the thickness of materials:



Beta particles can be used to measure the thickness of thin materials such as paper, cardboard or aluminium foil

- As a material moves above a beta source, the particles that are able to penetrate it can be monitored using a detector
- If the material gets thicker more particles will be absorbed, meaning that less will get through
- If the material gets thinner the opposite happens
- This allows the machine to make adjustments to keep the thickness of the material constant
- **Note**: Devices like this use beta radiation because it will be ***partially absorbed*** by the material
  If alpha particles were used all of them would be absorbed and none would get through
  If gamma were used almost all of it would get through and the detector would not be able to sense any difference if the thickness were to change

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Tracers

- Tracers are radioactive isotopes that can be added to some fluid so that the flow of that fluid can be monitored
- Tracers have numerous uses in both medicine and industry:
  - In medicine tracers can be added to the blood to check blood flow around the body and search for blockages (blood clots)
  - In industry tracers may be added into an oil pipeline in order to check for any leaks
- In all cases:
  - The amount used is kept to a minimum to reduce people's exposure to radiation
  - Isotopes are chosen that have short half-lives of around a few hours: long enough to carry out the procedure, but not so long that they cause long-term harm
  - Gamma radiation is used as it is highly penetrating (you can detect it) and low ionising (minimising harm)

Radiotherapy

- Radiotherapy is the name given to the treatment of cancer using radiation (Chemotherapy is treatment using chemicals)
- Although radiation can cause cancer, it is also highly effective at treating it
- Radiation can kill living cells. Some cells, such as bacteria and cancer cells, are more susceptible to radiation than others
- Beams of gamma rays are directed at the cancerous tumour
  (Gamma rays are used because they are able to penetrate the body, reaching the tumour)
- The beams are moved around to minimise harm to healthy tissue whilst still being aimed at the tumour

Sterilisation

- Medical instruments are sterilised by exposing them to gamma rays
- The gamma rays kill bacteria on the instruments and destroy viruses
- Gamma rays are far more effective at killing bacteria than either boiling water or chemical treatment and are able to penetrate the instruments reaching areas that may otherwise not be properly sterilised

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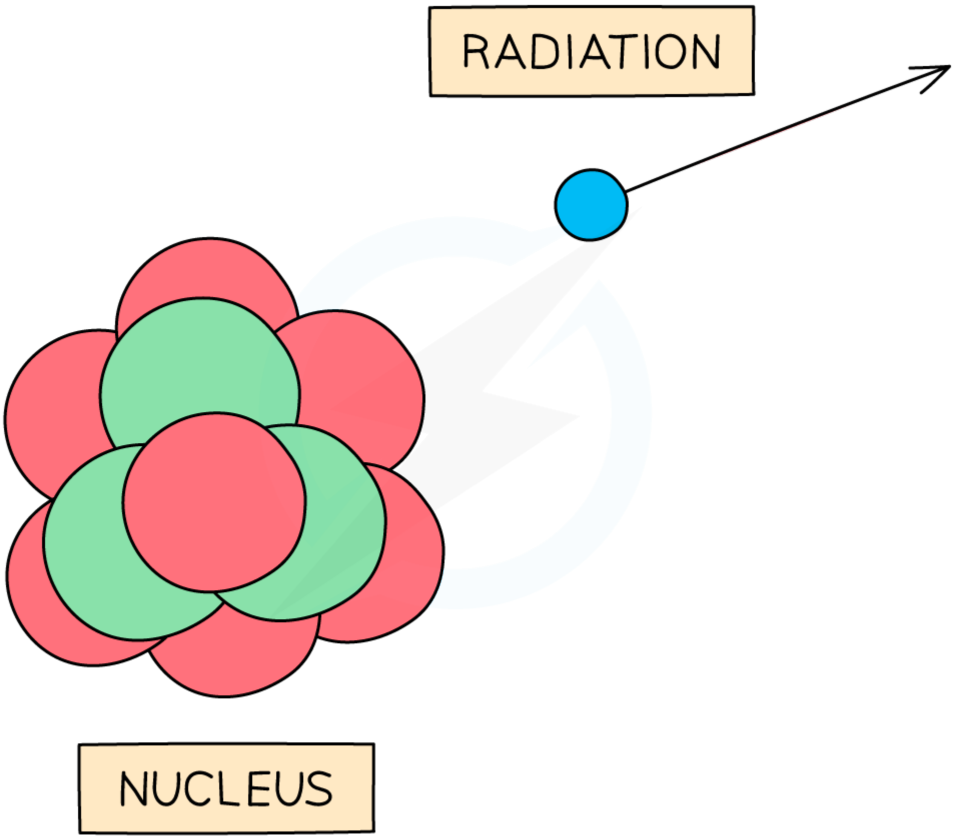
5.2.3 RADIOACTIVE DECAY

Radioactive Decay: Basics

- **Some isotopes are unstable** – usually because of their large size or because the number of protons and neutrons within them are out of balance
- As a result, these isotopes **will decay** – emitting little chunks (radiation) in order to reduce their size or bring them back into balance



Unstable nuclei decay by emitting high energy articles – radiation

- When an isotope emits radiation, the constitution of its nucleus (the number of protons and neutron) changes
- As a result, the isotope will change into a different element

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Alpha Emission

- An alpha particle consists of **2 protons and 2 neutrons** (It is emitted from large unstable nuclei)
- When an alpha particle is emitted from a nucleus:
    - The nucleus loses 2 protons:
      The proton (atomic) number decreases by 2
    - The nucleus loses 4 particles (nucleons) in total:
      The nucleon (mass) number decreases by 4
- Equation for alpha emission:

$$^{A}_{Z}X \rightarrow \, ^{A-4}_{Z-2}Y + \, ^{4}_{2}\alpha$$

$$\text{e.g.} \quad ^{212}_{84}Po \rightarrow \, ^{208}_{82}Y + \, ^{4}_{2}\alpha$$

- **Nuclear equations, just like chemical equations, balance**:
    - The sum of the upper (mass) numbers on the left of each equation should equal the sum on the right
    - The sum of the lower (atomic) numbers should also balance

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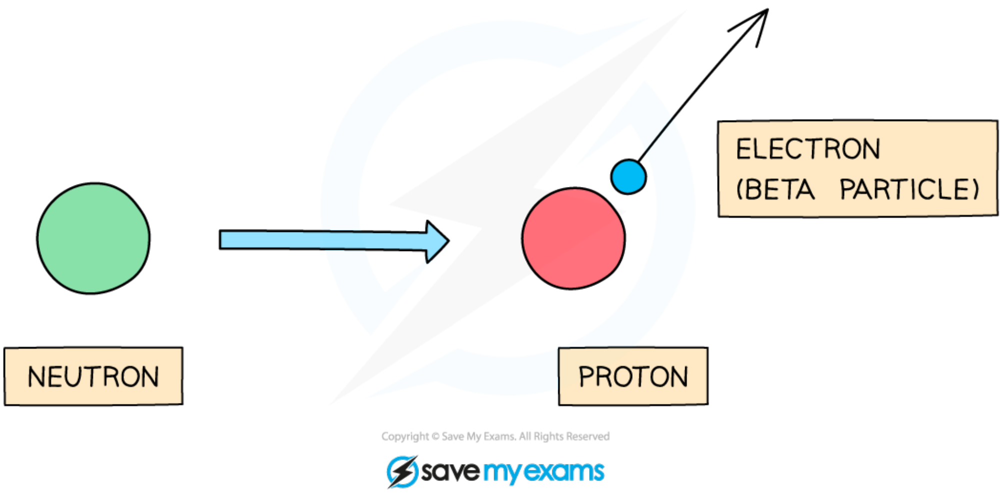
Beta Emission

- A beta particle is **a high energy electron** emitted from the nucleus
- It is emitted when **a neutron in the nucleus suddenly changes into a proton** – an electron is created in order to balance the positive charge of the proton
  (Note: The electron is created at the moment of decay – it is not present in the neutron beforehand)
- When a beta particle is emitted from a nucleus:
  - The number of protons in the nucleus increases by 1:
    The proton (atomic) number *increases* by 1
  - The total number of particles in the nucleus remains the same
    The nucleon (mass) number doesn't change



A beta particle is emitted when a neutron turns into a proton, emitting an electron

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- Equation for beta emission:

$$^{A}_{Z}X \longrightarrow \, ^{A}_{Z+1}Y + \, ^{0}_{-1}\beta$$

$$\text{e.g.} \quad ^{14}_{6}C \longrightarrow \, ^{14}_{7}N + \, ^{0}_{-1}\beta$$

- Note that the beta particle is given an atomic number of ***-1*** in the above examples
  This is because the atomic number is being used to measure charge in this case:
  Protons, being positive particles, have positive atomic numbers
  Electrons, being negative, have a negative number

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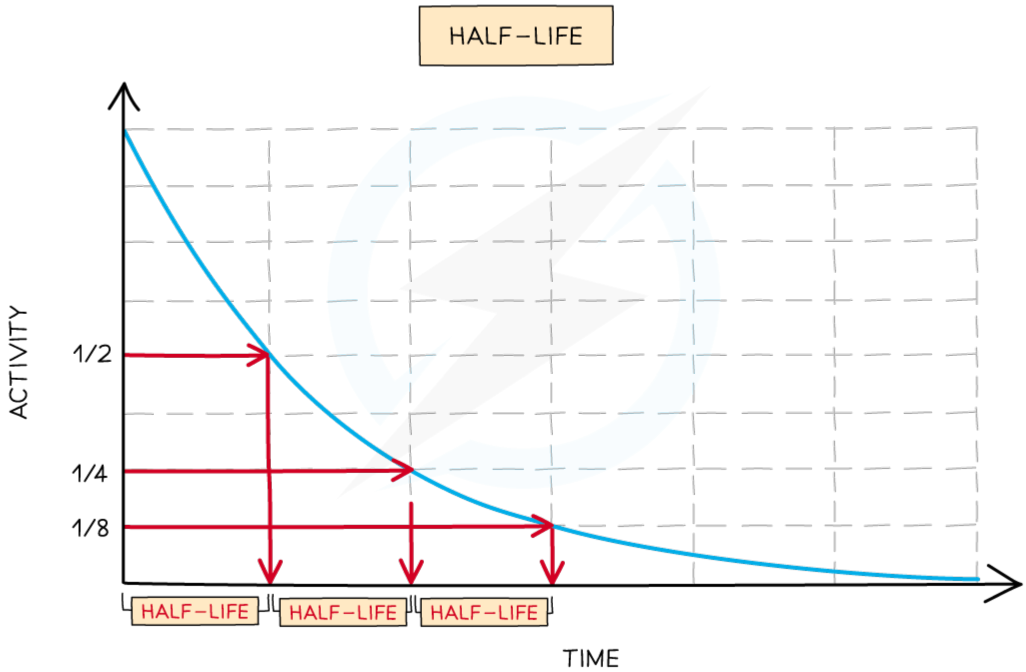
5.2.4 HALF-LIFE

Half-Life Basics

- As an isotope decays, the number of nuclei of that isotope that remain will decrease
- As a consequence of this, **the activity of that isotope will also decrease over time**
- **The half-life of an isotope is the time taken for the activity of that isotope (or the number of original nuclei) to drop to half of its initial value**
- Every time one half-life passes, the activity (and the number of nuclei) will fall by half



Graph showing the change in activity of an isotope over time and its radioactive half-life

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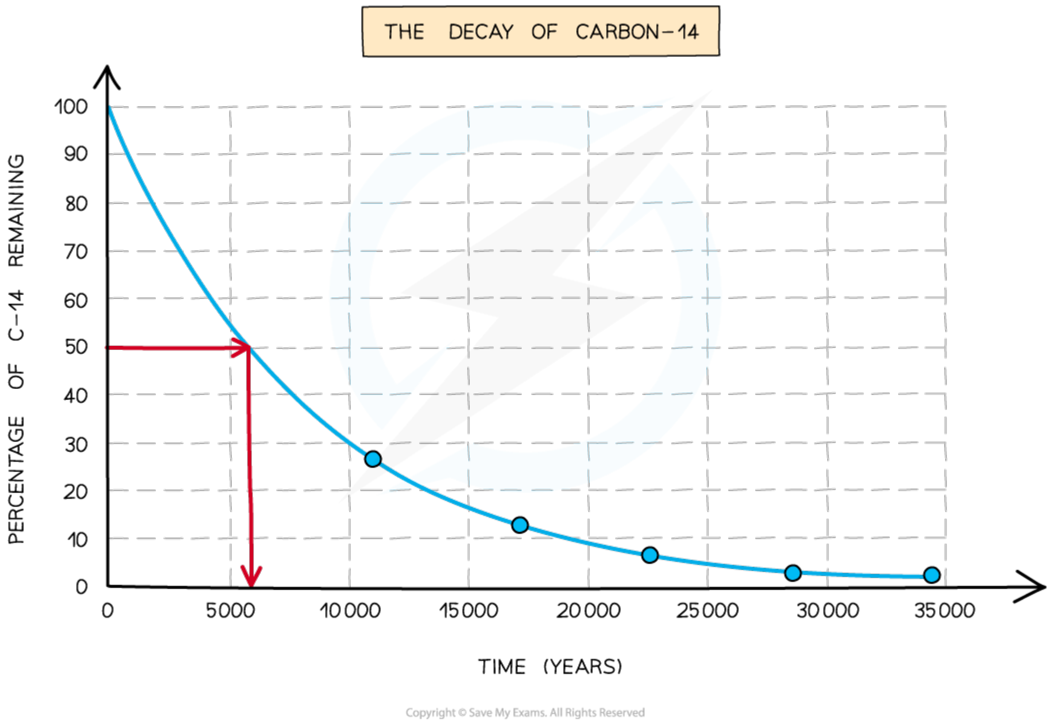
- Different isotopes have different half-lives and half-lives can vary from a fraction of a second to billions of years in length
- As mentioned above, every time one half-life passes the activity (and number of nuclei remaining) halves
  However, the activity (and number of nuclei) will never quite drop to zero

NUMBER OF HALF–LIVES	PROPORTION OF ISOTOPE REMAINING
0	1
1	1/2
2	1/4
3	1/8
4	1/16
...	...

Measuring Half-Life

To find the half-life of an isotope:

- If given some data showing how the activity (or number of nuclei) changes over time:
  - Plot a graph of this data (with time on the x-axis)
  - Add a smooth best fit curve (the curve should get closer to, but never quite reach, the x-axis)
  - Look at the original activity (where the line crosses the y-axis) and halve it
  - Go across from the halved value (on the y-axis) to the best fit curve, and then straight down to the x-axis
    (It's a good idea to draw lines showing this on your graph)
  - The point where you reach the x-axis should be the half-life



Use graphs like the one above to work out the half-life of an isotope

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- IF you are given just two pieces of data (along with a time) – say the initial and final activity of an isotope:
  - Start by figuring out how many times you have to halve the initial activity to get to the final activity
  - This number will be the number of half-lives that have passed
  - Divide the time by the number of half-lives to figure out the value of one half-life

Example:

An isotope has an initial activity of 120 Bq.
6 days later it's activity is 15 Bg.

The number of half-lives that have passed is:

120/2 = 60

60/2 = 30

30/2 = **15**

We had to halve 120 three times to get to 15, and so three half-lives have passed.

Therefore each half-life must be:

6 days/3 = 2 days

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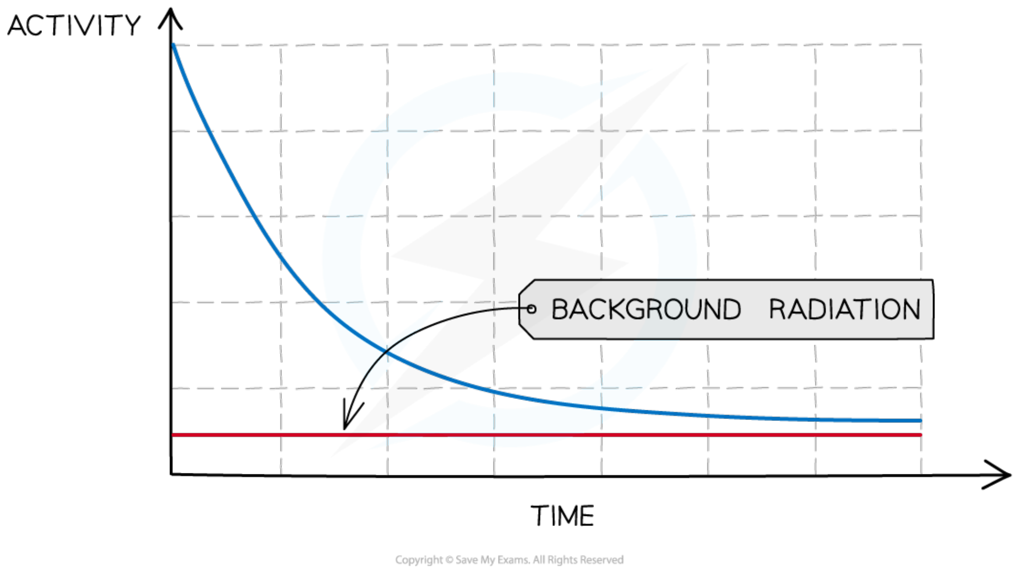
Background Radiation

- Background radiation is radiation that is always present in the environment around us
- As a consequence, whenever an experiment involving radiation is carried out, some of the radiation that is detected will be background radiation
- When carrying out experiments to measure half-life, the presence of background radiation must be taken into account



When measuring radioactive emissions, some of the detected radiation will be background

- To do this you must:
  - Start by measuring background radiation (with no sources present) – this is called your **background count**
  - Then carry out your experiment
  - Subtract the background count from each of your readings, in order to give a **corrected count**
  - The corrected count is your best estimate of the radiation emitted from the source, and should be used to measure its half-life

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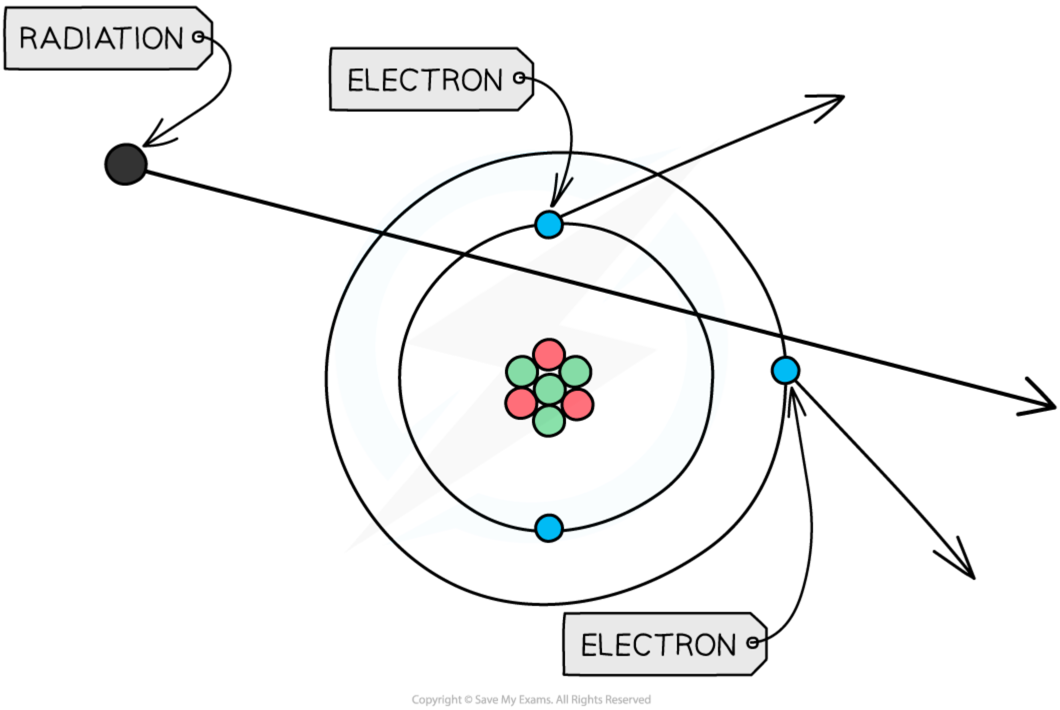
5.2.5 SAFETY PRECAUTIONS

The Dangers

- When radiation passes close to atoms the radiation can knock out electrons, ionising the atom



When radiation passes close to an atom it can knock electrons out of the atom, giving the atom a charge

- **Ionisation** can cause chemical changes in materials
- If these chemical changes occur in living cells it can damage the cell and:
  - **Cause mutations**
  - **Cause a cell to become cancerous**
  - **Kill the cell**

5. Atomic Physics

YOUR NOTES
⬇

Safety

Radioactivity warning sign

- The risks associated with handling radioactive sources can be minimised by following a few simple procedures:
  - Store the sources in lead-lined boxes and keep at a distance from people
  - Minimise the amount of time you handle sources for and return them to their boxes as soon as you have finished using them
  - During use, keep yourself (and other people) as far from the sources as feasible. When handling the sources do so at arm's length, using a pair of tongs

(**Note**: When using tongs, gloves and safety specs are usually unnecessary when handling radioactive materials, unless there is a risk of the material leaking on to things)

5. Atomic Physics

YOUR NOTES

Exam Question: Easy

A student carried out an experiment to find the half-life of a radioactive substance. Their results are shown in the table below.

time (seconds)	count-rate from source (counts per second)
0	300
20	200
40	150
60	100
80	75

What is the half-life of this substance?

A 20 seconds.

B 40 seconds

C 60 seconds

D 80 seconds

Exam Question: Medium

There are three main types of radiation which may be emitted during radioactive decay: α-particles, β-particles and γ-rays.

Which of the following statements about these types of emissions is **true**?

A α-particles are the only type of radiation to have a charge.

B γ-rays are stopped by a sheet of paper.

C β-particles have the greatest ionising effect.

D γ-rays are the most penetrating

5. Atomic Physics

YOUR NOTES

Exam Question: Hard

A radioactive substance has a half-life of 4 days.

It is currently emitting 8000 β-particles per minute.

How many β-particles will it emit per minute after 12 days?

A 4000

B 2000

C 1000

D 667

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